

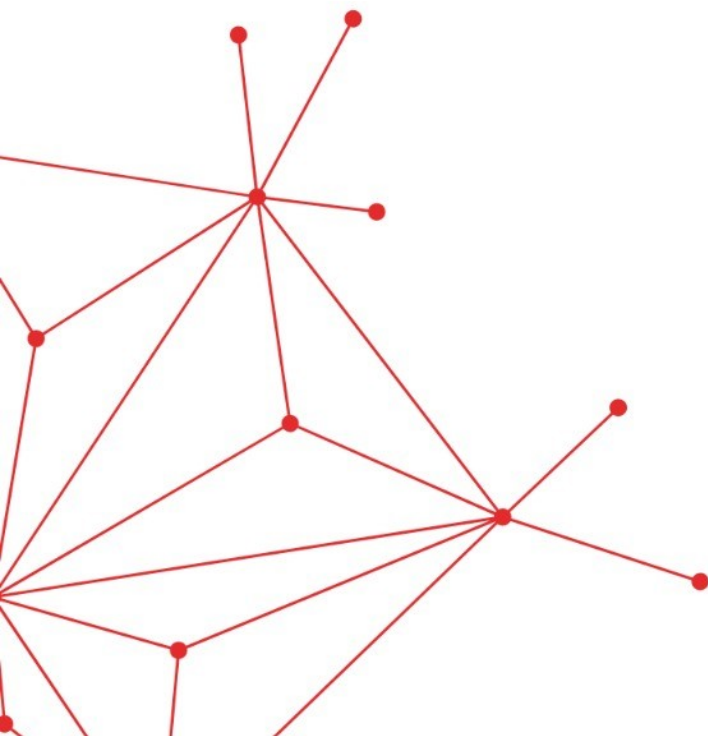


Deep Dive

Into

Advanced Statistics

Part-1: Probability



Introduction to Probability

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1 Introduction to Probability

1.1 Why Probability?

Life frequently presents us with uncertain events, urging us to assess unpredictable outcomes before making decisions.

Examples:

- Imagine preparing for an exam: What are the chances of passing or failing based on your level of preparation?
- Consider launching a new business: What are the odds of success or the potential risks of failure?

The answers to these questions rely on understanding and navigating uncertainty effectively.

Randomness is an integral aspect of our lives, yet humans often find it challenging to interpret and understand. By learning to comprehend randomness, we can leverage probability as a powerful tool to quantify and make sense of it. Probability provides us with the language and methodology to measure the uncertainty inherent in various events.

Examples:

- Flipping a coin: When you toss a coin, there is an equal chance of landing on heads or tails. Although the outcome appears random, probability indicates that the likelihood of each result is 50%.
- Rolling a die: Each face of a standard six-sided die has an identical chance of being rolled. Probability helps quantify this uncertainty by assigning a 1-in-6 chance to each number.
- Drawing cards from a deck: The act of drawing a specific card from a well-shuffled deck might seem random. Probability clarifies this by showing that there is a 1-in-52 chance of selecting any particular card.

These examples emphasize how randomness influences everyday situations, while probability offers a systematic framework to measure and better understand this uncertainty.

The ability to recognize and reason about randomness is essential. It helps not only in statistical analyses but also in making informed decisions in everyday life. For example, meteorologists use probability to predict the likelihood of rain, and investors evaluate risks when deciding where to put their money. These are skills that can dramatically improve our ability to interpret the events around us.

The ability to identify and reason about randomness is a crucial skill. It extends far beyond statistical analyses and plays a significant role in helping us make informed decisions in various aspects of our daily lives.

Examples:

- Weather forecasting: Meteorologists use probability to estimate the likelihood of rain, storms, or sunny days, helping people plan their outdoor activities or decide whether to carry an umbrella.
- Investing in the stock market: Investors assess the risks and potential returns of various stocks, relying on probability to make educated choices about where to allocate their funds.
- Medical diagnostics: Doctors use probability to evaluate the likelihood of a patient having a particular condition based on symptoms and test results, ensuring more accurate diagnoses and treatments.
- Sports and games: Coaches and players strategize using the probabilities of success for different plays, while fans might use odds to predict match outcomes.
- Traffic and commuting: People often estimate the chances of encountering traffic jams or delays and choose their routes or departure times accordingly.
- Elections and polling: Probability is used to interpret opinion polls and predict outcomes, giving insights into public sentiment and potential results.
- Education and exams: Students often assess the likelihood of certain topics appearing on exams and adjust their study strategies accordingly.

By understanding and reasoning about randomness, we gain the tools to make thoughtful decisions and better interpret the unpredictable nature of everyday situations.

The more we reflect, the more we notice that randomness is everywhere—in traffic patterns, medical diagnoses, or even the outcomes of rolling dice. Our vocabulary, rich with terms like "uncertainty," "risk," "chance," and "likelihood," reflects how deeply randomness is woven into our daily lives.

1.2 What is probability?

Probability is the quantification of randomness, providing a systematic way to measure how likely an event is to occur. It reflects the likelihood of outcomes based on observation and analysis. Probability, according to the classical approach, measures the likelihood of an event happening.

Formula:

If there are n possible outcomes and they are equally likely, the probability of an event (A) is given by:

$$P(A) = \frac{\text{Number of favorable outcomes in } A}{\text{Total number of possible outcomes}}$$

- The numerator represents the event (A), the favorable outcomes to the experiment, or the size of the event.
- The denominator represents the sample space, the total number of possible outcomes, or the size of the sample space.

Examples:

- Flipping a fair coin: The probability of landing heads is ($P(\text{Heads}) = 0.5$ ($1/2$), since there is 1 favorable outcome (heads) out of 2 possible outcomes (heads or tails).
- Rolling a six-sided die: The probability of rolling a 4 is ($P(\text{Rolling a 4}) = 0.167$ ($1/6$), as there is 1 favorable outcome (4) out of 6 possible outcomes (1, 2, 3, 4, 5, and 6).
- Drawing a red card: A standard deck of playing cards contains 52 cards, half of which (26) are red (hearts and diamonds). The probability of drawing a red card is:

$$P(\text{Red Card}) = \frac{\text{Number of red cards}}{\text{Total cards}} = \frac{26}{52} = \frac{1}{2}$$

- Drawing a King: There are 4 Kings in a deck (one for each suit). The probability of drawing a King is:

$$P(\text{King}) = \frac{\text{Number of Kings}}{\text{Total cards}} = \frac{4}{52} = \frac{1}{13}$$

- Drawing a heart: There are 13 cards in each suit (clubs, diamonds, hearts, and spades). The probability of drawing a heart is:

$$P(\text{Heart}) = \frac{\text{Number of hearts}}{\text{Total cards}} = \frac{13}{52} = \frac{1}{4}$$

- Drawing an Ace: There are 4 Aces in a deck. The probability of drawing an Ace is:

$$P(\text{Ace}) = \frac{\text{Number of Aces}}{\text{Total cards}} = \frac{4}{52} = \frac{1}{13}$$

- Drawing a face card (King, Queen, Jack): A deck has 12 face cards (3 for each suit). The probability of drawing a face card is:

$$P(\text{Face Card}) = \frac{\text{Number of face cards}}{\text{Total cards}} = \frac{12}{52} = \frac{3}{13}$$

- Drawing a card with a number (2-10): There are 36 numbered cards in the deck (9 cards for each suit). The probability of drawing a numbered card is:

$$P(\text{Numbered Card}) = \frac{\text{Number of numbered cards}}{\text{Total cards}} = \frac{36}{52} = \frac{9}{13}$$

In probability, an experiment refers to any process that produces an outcome with inherent uncertainty. By using this framework, probability offers a structured way to reason about and understand random events.

1.2.1 Interpreting Probability Values

Probability is expressed as a value ranging from 0 to 1. These values help quantify the likelihood of an event occurring:

- A probability of 0 means the event is absolutely certain not to happen. For instance, if you roll a standard six-sided die, the probability of rolling a 7 is 0, as it's impossible (a die only has numbers from 1 to 6).
- A probability of 1 signifies the event is absolutely certain to occur. For example, when flipping a coin, the probability of landing on either heads or tails is 1, as one of these outcomes is guaranteed.
- Higher probability values (closer to 1) indicate a greater likelihood of the event occurring. For instance, if a weather forecast predicts a 90% chance of rain, it's highly likely (though not guaranteed) that it will rain.
- Conversely, lower probability values (closer to 0) indicate a lower chance of the event occurring. For example, a 10% chance of rain implies it's unlikely to rain, but there's still a small possibility.

1.3 Random experiment

A random experiment is a process or procedure that produces outcomes, where the exact outcome cannot be predicted in advance, even though the set of possible outcomes is known.

For example, hitting shuffle on your music player is a random experiment. You are aware of the possible outcomes—the next song could be any track from your music library. However, you cannot predict exactly which song will play next. This lack of certainty about the specific result, while knowing the range of outcomes, is what defines the randomness in such an experiment.

By observing or conducting experiments like this, we can gather insights into the phenomenon, analyze patterns, and explore probabilities associated with the outcomes. Random experiments like this example are widely used in statistics, probability studies, and real-life scenarios.

Examples

1. Flipping a fair coin

This random experiment involves tossing a coin with two distinct faces—heads and tails. The coin is flipped and allowed to land on a surface, where it settles on one of its sides. The outcome of each flip is unpredictable and can result in either heads or tails. Both outcomes are equally likely, making the probability of each 0.5 ($\frac{1}{2}$).

2. Rolling a six-sided die

This random experiment involves rolling a standard die with six numbered faces (1 to 6). The die is rolled and comes to rest, revealing one number on its top face. Each outcome is equally likely, so the probability of any specific number (e.g., rolling a 4) is $(1/6)$ or approximately 0.167.

3. Drawing a card from a deck

This random experiment involves randomly selecting one card from a standard deck of 52 playing cards. The deck is thoroughly shuffled to ensure each card has an equal chance of being drawn. The possible outcomes include any one of the 52 unique cards in the deck, with the probability of drawing a specific card (e.g., Ace of Spades) being $(1/52)$. The experiment of drawing a card from a standard deck can be classified based on different categories of outcomes. This experiment can be classified based on various aspects:

1. By color: Red cards (hearts and diamonds) or black cards (clubs and spades).

Red cards: 26 cards (hearts and diamonds).

Black cards: 26 cards (clubs and spades).

Probability of drawing a red card is: $(26/52 \sim \frac{1}{2})$

2. By suit: Drawing from one of the four suits.

Hearts: 13 cards.

Diamonds: 13 cards.

Clubs: 13 cards.

Spades: 13 cards.

Probability of drawing a heart is $(13/52 = 1/4)$

3. By rank: Selecting specific ranks like an Ace or a numbered card.

Specific rank: Each rank (e.g., Ace, 2, 3, etc.) has 4 cards (one in each suit).

Probability of drawing an Ace is $(4/52 = 1/13)$

4. By Face Cards: There are 12 face cards (King, Queen, Jack) across all four suits.

Non-face cards: 40 cards (Ace and numbered cards).

The Probability of drawing a face card is $(12/52 = 3/13)$

5. By Numbered and Non-Numbered Cards:

Numbered cards: 36 cards (2 to 10 across all suits).

Non-numbered cards: 16 cards (Ace, King, Queen, Jack across all suits).

The Probability of drawing a numbered card is $(36/52 = 9/13)$.

1.3.1 Trial

A trial refers to each individual repetition of an experiment, along with the recording of its outcome. In other words, a trial is a single instance of conducting an experiment to observe and document the result. When multiple trials are performed, they collectively constitute the complete experiment. The purpose of conducting numerous trials is to gather sufficient data for analysis and to draw meaningful conclusions.

Examples

1. Flipping a coin: Each flip of the coin, where you record whether it lands on heads or tails, constitutes a trial. Flipping the coin 10 times and noting the outcomes would be an experiment consisting of 10 trials.
2. Rolling a die: Rolling a six-sided die and recording the result (e.g., the number that lands face up) is a single trial. Rolling the die 20 times and documenting each result would make up an experiment with 20 trials.
3. Drawing a card: Drawing a card from a shuffled deck and noting whether it is red or black constitutes a trial. Repeating this process 30 times and recording the outcome of each draw forms an experiment with 30 trials.

1.3.2 Outcome

An outcome is the result of a single trial conducted in a probability experiment. It represents one specific result that can occur when the random process is performed.

Examples:

1. In a coin-tossing experiment, possible outcomes include heads or tails.
2. When rolling a six-sided die, the outcomes are the numbers 1, 2, 3, 4, 5, and 6.
3. Drawing a card from a shuffled deck can result in outcomes like Ace of Spades or 10 of Hearts.
4. In a spinner experiment with sections labeled A, B, and C, the outcomes are A, B, or C.

1.3.3 Sample space

A sample space refers to the complete set of all possible outcomes of a random experiment or phenomenon. It serves as a framework for analyzing the randomness and provides a comprehensive list of every result that could occur during a trial or experiment. Each individual result in the sample space is known as an elementary outcome, while a combination of outcomes is called an event.

Examples:

1. **Flipping a coin:** The sample space consists of the two possible outcomes: {Heads, Tails}.
2. **Rolling a six-sided die:** The sample space contains all six outcomes: { 1, 2, 3, 4, 5, 6 }.
3. **Drawing a card from a deck:** The sample space consists of all 52 cards in a standard deck, such as {Ace of Hearts, 2 of Diamonds, Queen of Spades, etc.}.
4. **Rolling two dice:** The sample space includes all 36 possible combinations of numbers from two dice rolls, such as { (1,1), (1,2), (2,3), ... (6,6) }.

1.3.4 Event

An event is a subset of the sample space in a random experiment. It can be a specific outcome or a combination of multiple outcomes. In probability, every event is associated with a probability that quantifies the likelihood of the event occurring. This probability can be calculated through experiments that observe and record the outcomes. Essentially, an event is defined as either a single result or a group of possible results within the sample space.

Example:

In the experiment of tossing a six-sided die, observing a specific number (e.g., rolling a 4) is an event. Similarly, observing an even number (e.g., rolling 2, 4, or 6) is an event that consists of multiple outcomes.

1.3.4.1 Simple/Elementary event

A simple event is an individual outcome observed during a single trial of a random experiment. It represents one distinct result within the sample space and cannot be broken down further.

OR

Events that cannot be decomposed or broken down into other events are called elementary events.

Example:

In the experiment of tossing a six-sided die, the six simple events are:

- Event (E_1): Observe a 1
- Event (E_2): Observe a 2
- Event (E_3): Observe a 3
- Event (E_4): Observe a 4
- Event (E_5): Observe a 5
- Event (E_6): Observe a 6

Each of these simple events corresponds to a unique outcome of the die toss.

1.3.4.2 Compound event

A compound event is a collection of simple events grouped together based on specific criteria. It includes multiple possible outcomes and is represented as a subset of the sample space.

Example:

For the die-tossing experiment:

- Event A: Observing an odd number. This event includes the outcomes ($A = \{ E_1, E_3, E_5 \}$), corresponding to rolling 1, 3, or 5.
- Event B: Observing a number less than 4. This event consists of outcomes ($B = \{ E_1, E_2, E_3 \}$), representing rolling 1, 2, or 3.

The probability of a compound event can be calculated as the sum of the probabilities of the simple events contained within it. For example, the probability of observing an odd number (Event A) is the sum of the probabilities of rolling 1, 3, and 5.

1.3.5 Defining New Events

Once we have defined some events, there are various ways to combine and manipulate them to create new events. Venn diagrams are an essential tool for visualizing and calculating probabilities, enabling us to combine and manipulate events to create new scenarios. Below are formal definitions for constructing new events:

Complement of an Event

The complement of an event (A), denoted by (A^c), consists of all outcomes in the sample space (S) that are not in (A). The complement represents the event “not (A).”

Example: If (A) is the event "rolling a number greater than 4" in a dice experiment, then (A^c) is the event "rolling a number less than or equal to 4."

Union of Two Events

The union of two events (A) and (B), denoted as $A \cup B$, includes all outcomes that are in at least one of the two events. This means that the outcome can be in (A), in (B), or in both. The union is commonly referred to as "either (A) or (B)."

Example: In a dice experiment, let (A) represent rolling an even number ($\{2, 4, 6\}$) and (B) represent rolling a number greater than 4 ($\{5, 6\}$). The union $A \cup B$ is $\{2, 4, 5, 6\}$.

Intersection of Two Events

The intersection of two events (A) and (B), denoted as $A \cap B$, consists of all outcomes that are in both events. The intersection identifies shared outcomes.

Example: Using the same dice experiment, let (A) represent rolling an even number ($\{2, 4, 6\}$) and (B) represent rolling a number greater than 4 ($\{5, 6\}$). The intersection $A \cap B$ is $\{6\}$, as 6 is the only outcome common to both events.

Extending Unions and Intersections

These operations can be generalized to combine more than two events. For example:

- The union of three events (A), (B), and (C), denoted $A \cup B \cup C$, includes all outcomes that are in at least one of the three events.
- The intersection of three events (A), (B), and (C), denoted $A \cap B \cap C$, includes outcomes that are common to all three events.

Examples on classical approach to probability

1. What is the probability of {H} by tossing a single coin?

In the experiment of tossing a single coin, the **event** {H} represents the outcome of the coin landing on heads. Tossing a coin is a **random experiment** because its outcome is uncertain, but the set of all possible outcomes, called the **sample space**, is known. In this case: $S = \{H, T\}$

Each outcome is equally likely. The **event** {H} corresponds to the single favorable outcome: the coin landing on heads. Using the formula for probability:

$$P(H) = \frac{\text{Number of favorable outcomes for heads}}{\text{Total number of possible outcomes}} = \frac{1}{2} = 0.5$$

Thus, the probability of {H} is **0.5**, or 50%.

2. What is the probability of {T} by tossing a single coin?

The **event {T}** refers to the outcome of the coin landing on tails. Similar to the first case, the **sample space** is: $S = \{H, T\}$]

The event {T} corresponds to the single favorable outcome: the coin landing on tails. The calculation for probability is:

$$P(T) = \frac{\text{Number of favorable outcomes for tails}}{\text{Total number of possible outcomes}} = \frac{1}{2} = 0.5$$

Hence, the probability of {T} is also **0.5**, or 50%.

3. What is the probability of {} by tossing a single coin?

The event {} represents the **empty set**, meaning no outcome occurs. In a coin toss, this event is **impossible** because the coin must land on either heads (H) or tails (T). The **sample space** remains: $S = \{H, T\}$

Since there are no favorable outcomes for the event {}, its probability is: $[P(\{\}) = 0]$

Thus, the probability of {} is **0**, indicating an impossible event.

4. What is the probability of {H, T} by tossing a single coin?

The **event {H, T}** includes both possible outcomes of the coin toss. This event represents a **certain event**, meaning the coin must either land on heads or tails. The **sample space** is: $S = \{H, T\}$

Since both heads (H) and tails (T) are favorable outcomes for the event {H, T}, its probability is:

$$P(H, T) = \frac{\text{Number of favorable outcomes (H or T)}}{\text{Total number of possible outcomes}} = \frac{2}{2} = 1$$

Hence, the probability of {H, T} is **1**, or 100%, indicating a certain event.

5. What is the probability of two heads by tossing two coins?

- **Random Experiment:** Tossing two coins is a random experiment, where the outcome of each toss is uncertain.
- **Sample Space:** The possible outcomes when tossing two coins (each coin landing on heads or tails) form the sample space: {HH, HT, TH, TT}
- **Event:** {HH}, The event {HH} represents both coins landing on heads.

- **Favorable outcomes:** 1 (HH).
- **Total outcomes:** 4 (HH, HT, TH, TT).

$$P(\text{HH}) = \frac{\text{Number of favorable outcomes for event } A}{\text{Total number of outcomes in the sample space}} = \frac{1}{4} = 0.25, (25\%)$$

6. What is the probability of two tails by tossing two coins?

- **Random Experiment:** Tossing two coins is a random experiment, where the outcome of each toss is uncertain.
- **Sample Space:** The possible outcomes when tossing two coins (each coin landing on heads or tails) form the sample space: {HH, HT, TH, TT}
- **Event:** {TT}.
- **Favorable outcomes:** 1 (TT).
- **Total outcomes:** 4 (HH, HT, TH, TT).

$$P(\text{TT}) = \frac{\text{Number of favorable outcomes for event } A}{\text{Total number of outcomes in the sample space}} = \frac{1}{4} = 0.25, (25\%)$$

7. What is the probability of head and tails by tossing two coins?

- **Random Experiment:** Tossing two coins is a random experiment, where the outcome of each toss is uncertain.
- **Sample Space:** The possible outcomes when tossing two coins (each coin landing on heads or tails) form the sample space: {HH, HT, TH, TT}
- **Event:** {HT, TH}, The event {HT, TH} represents one coin landing on heads and the other landing on tails.
- **Favorable outcomes:** 2 (HT, TH).
- **Total outcomes:** 4 (HH, HT, TH, TT).

$$P(\text{HT, TH}) = \frac{\text{Number of favorable outcomes for event } A}{\text{Total number of outcomes in the sample space}} = \frac{2}{4} = 0.5, (50\%)$$

8. What is the probability of head on one coin and tails on another coin?

- **Random Experiment:** Tossing two coins is a random experiment where the outcome of each coin flip is uncertain.

- **Sample Space:** When tossing two coins, the sample space includes all possible combinations: $S = \{HH, HT, TH, TT\}$
- **Event:** The coin toss results in **HT or TH**, where one coin lands on heads and the other lands on tails.
- **Favorable outcomes:** $\{HT, TH\}$ (2 outcomes).
- **Total outcomes:** 4 (HH, HT, TH, TT).
- **Probability:**

$$P(HT, TH) = \frac{\text{Number of favorable outcomes for event } A}{\text{Total number of outcomes in the sample space}} = \frac{2}{4} = 0.5, (50\%)$$

9. What is the probability of showing 1 on its face by rolling a single die?

- **Random Experiment:** Rolling a single six-sided die is a random experiment, where the number appearing on the top face is uncertain.
- **Sample Space:** The sample space for rolling a die is: $S = \{1, 2, 3, 4, 5, 6\}$
- **Event:** Rolling the number 1.
- **Favorable outcomes:** $\{1\}$ (1 outcome).
- **Total outcomes:** 6 (1, 2, 3, 4, 5, 6).
- **Probability:**

$$P(1) = \frac{\text{Number of favorable outcomes for event } A}{\text{Total number of outcomes in the sample space}} = \frac{1}{6} \approx 0.167, (16.7\%)$$

10. What is the probability of showing {1, 2} on its face by rolling a single die?

- **Random Experiment:** Rolling a single six-sided die is a random experiment, where the number appearing on the top face is uncertain.
- **Sample Space:** The sample space for rolling a die is: $S = \{1, 2, 3, 4, 5, 6\}$
- **Event:** Rolling the numbers 1 or 2.
- **Favorable outcomes:** $\{1, 2\}$ (2 outcomes).
- **Total outcomes:** 6 (1, 2, 3, 4, 5, 6).
- **Probability:**

$$P(1, 2) = \frac{\text{Number of favorable outcomes for event } A}{\text{Total number of outcomes in the sample space}} = \frac{2}{6} = \frac{1}{3} \approx 0.333, (33.3\%)$$

11. What is the probability of showing an even number on its face by rolling a single die?

- **Random Experiment:** Rolling a single six-sided die is a random experiment, where the number appearing on the top face is uncertain.
- **Sample Space:** The sample space for rolling a die is: $S = \{1, 2, 3, 4, 5, 6\}$
- **Event:** Rolling an even number.
- **Favorable outcomes:** $\{2, 4, 6\}$ (3 outcomes).
- **Total outcomes:** 6 (1, 2, 3, 4, 5, 6).
- **Probability:**

$$P(\text{Even}) = \frac{\text{Number of favorable outcomes for event } A}{\text{Total number of outcomes in the sample space}} = \frac{3}{6} = \frac{1}{2} = 0.5, (50\%)$$

12. What is the probability of showing an odd number on its face by rolling a single die?

- **Random Experiment:** Rolling a single six-sided die is a random experiment, where the number appearing on the top face is uncertain.
- **Sample Space:** The sample space for rolling a die is: $S = \{1, 2, 3, 4, 5, 6\}$
- **Event:** Rolling an odd number.
- **Favorable outcomes:** $\{1, 3, 5\}$ (3 outcomes).
- **Total outcomes:** 6 (1, 2, 3, 4, 5, 6).
- **Probability:**

$$P(\text{Odd}) = \frac{\text{Number of favorable outcomes for event } A}{\text{Total number of outcomes in the sample space}} = \frac{3}{6} = \frac{1}{2} = 0.5, (50\%)$$

13. What is the probability of showing an even number and an odd number on its face by rolling a single die?

- **Random Experiment:** Rolling a single six-sided die is a random experiment, where the number appearing on the top face is uncertain.
- **Sample Space:** The sample space for rolling a die is: $S = \{1, 2, 3, 4, 5, 6\}$
- **Event:** Rolling a number that is both even and odd.
- **Favorable outcomes:** None (0 outcomes).
- **Total outcomes:** 6 (1, 2, 3, 4, 5, 6).
- **Probability:**

$$P(\text{Even and Odd}) = \frac{\text{Number of favorable outcomes for event } A}{\text{Total number of outcomes in the sample space}} = 0$$

14. What is the probability of showing an even number or an odd number on its face by rolling a single die?

- **Random Experiment:** Rolling a single six-sided die is a random experiment, where the number appearing on the top face is uncertain.
- **Sample Space:** The sample space for rolling a die is: $S = \{1, 2, 3, 4, 5, 6\}$
- **Event:** Rolling a number that is either even or odd.
- **Favorable outcomes:** All outcomes $\{1, 2, 3, 4, 5, 6\}$ (6 outcomes).
- **Total outcomes:** 6 (1, 2, 3, 4, 5, 6).
- **Probability:**

$$P(\text{Even or Odd}) = \frac{\text{Number of favorable outcomes for event } A}{\text{Total number of outcomes in the sample space}} = \frac{6}{6} = 1$$

15. What is the probability of showing a number greater than 3 on its face by rolling a single die?

- **Random Experiment:** Rolling a single six-sided die is a random experiment, where the number appearing on the top face is uncertain.
- **Sample Space:** The sample space for rolling a die is: $S = \{1, 2, 3, 4, 5, 6\}$
- **Event:** Rolling a number greater than 3.
- **Favorable outcomes:** $\{4, 5, 6\}$ (3 outcomes).
- **Total outcomes:** 6 (1, 2, 3, 4, 5, 6).
- **Probability:**

$$P(\text{Number } > 3) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes in the sample space}} = \frac{3}{6} = \frac{1}{2} = 0.5, (\text{or } 50\%).$$

16. What is the probability of showing a number less than 4 on its face by rolling a single die?

- **Random Experiment:** Rolling a single six-sided die is a random experiment, where the number appearing on the top face is uncertain.
- **Sample Space:** The sample space for rolling a die is: $S = \{1, 2, 3, 4, 5, 6\}$
- **Event:** Rolling a number less than 4.
- **Favorable outcomes:** $\{1, 2, 3\}$ (3 outcomes).

- **Total outcomes:** 6 (1, 2, 3, 4, 5, 6).
- **Probability**

$$P(\leq 4) = \frac{\text{Number of favorable outcomes for event } A}{\text{Total number of outcomes in the sample space}} = \frac{3}{6} = \frac{1}{2} = 0.5, (50\%)$$

17. What is the probability of showing an even number and a number less than 4 on its face by rolling a single die?

- **Random Experiment:** Rolling a single six-sided die is a random experiment, where the number appearing on the top face is uncertain.
- **Sample Space:** The sample space for rolling a die is: $S = \{1, 2, 3, 4, 5, 6\}$
- **Event:** Rolling an even number less than 4.
- **Favorable outcomes:** $\{2\}$ (1 outcome).
- **Total outcomes:** 6 (1, 2, 3, 4, 5, 6).
- **Probability**

$$P(\text{Even and } \leq 4) = \frac{\text{Number of favorable outcomes for event } A}{\text{Total number of outcomes in the sample space}} = \frac{1}{6} \approx 0.167, (16.7\%)$$

18. What is the probability of showing an even number and a number less than 5 on its face by rolling a single die?

- **Random Experiment:** Rolling a single six-sided die is a random experiment, where the number appearing on the top face is uncertain.
- **Sample Space:** The sample space for rolling a die is: $S = \{1, 2, 3, 4, 5, 6\}$
- **Event:** Rolling an even number less than 5.
- **Favorable outcomes:** $\{2, 4\}$ (2 outcomes).
- **Total outcomes:** 6 (1, 2, 3, 4, 5, 6).
- **Probability**

$$P(\text{Even and } \leq 5) = \frac{\text{Number of favorable outcomes for event } A}{\text{Total number of outcomes in the sample space}} = \frac{2}{6} = \frac{1}{3} \approx 0.333, (33.3\%)$$

19. What is the probability of showing an odd number and a number less than 4 on its face by rolling a single die?

- **Random Experiment:** Rolling a single six-sided die is a random experiment, where the number appearing on the top face is uncertain.
- **Sample Space:** The sample space for rolling a die is: $S = \{1, 2, 3, 4, 5, 6\}$
- **Event:** Rolling an odd number less than 4.
- **Favorable outcomes:** $\{1, 3\}$ (2 outcomes).
- **Total outcomes:** 6 (1, 2, 3, 4, 5, 6).
- **Probability:**

$$P(\text{Odd and } <4) = \frac{\text{Number of favorable outcomes for event } A}{\text{Total number of outcomes in the sample space}} = \frac{2}{6} = \frac{1}{3} \approx 0.333, (33.3\%)$$

20. What is the probability of showing an odd number and a number less than 3 on its face by rolling a single die?

- **Random Experiment:** Rolling a single six-sided die is a random experiment, where the number appearing on the top face is uncertain.
- **Sample Space:** The sample space for rolling a die is: $S = \{1, 2, 3, 4, 5, 6\}$
- **Event:** Rolling an odd number less than 3.
- **Favorable outcomes:** $\{1\}$ (1 outcome).
- **Total outcomes:** 6 (1, 2, 3, 4, 5, 6).
- **Probability:**

$$P(\text{Odd and } <3) = \frac{\text{Number of favorable outcomes for event } A}{\text{Total number of outcomes in the sample space}} = \frac{1}{6} \approx 0.167, (16.7\%)$$

1.4 Axioms of Probability

Probability provides a structured framework to quantify uncertainty and measure the likelihood of events. Formal probability employs mathematical notation to express foundational principles more concisely. Given an experiment with a sample space (S), probability aims to assign a numerical value ($P(A)$) to each event (A). This value reflects our intuitive understanding of likelihood while ensuring mathematical consistency. To achieve this, probability assignments adhere to fundamental rules known as the **axioms of probability**.

These axioms, based on commonsense principles, provide the foundation for probability theory and align with intuitively true concepts, such as proportions in repeated trials of random experiments. Together, they guarantee that our probability assignments make sense and work coherently for real-world and theoretical applications.

1.4.1 The Three Axioms of Probability

1. **Probability Range Rule:** For any event (A), the probability (P(A)) satisfies: $0 \leq P(A) \leq 1$ This means that the likelihood of an event is always between 0 (impossible event) and 1 (certain event).
2. **Sample Space Axiom:** If (S) is the sample space of a random experiment, then the probability of the sample space (S) is: $P(S) = 1$ This reflects the certainty that one of the possible outcomes in the sample space will occur.

Additivity for Mutually Exclusive Events: If (A) and (B) are **mutually exclusive events** (i.e., they cannot occur simultaneously), then: $P(A \cup B) = P(A) + P(B)$ For more than two mutually exclusive events $A_1, A_2, \dots$, the rule generalizes as: $P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$

1.4.2 Special Cases of axioms

1. **Complement Rule:** The complement of an event (A), denoted by $(A^c) = 1 - P(A)$. It also follows that
$$P(A) = 1 - P(A^c)$$
2. **Equally Likely Outcomes:** If all outcomes in the sample space (S) are equally likely, the probability of an event (A) is given by

$$P(A) = \frac{\text{Number of outcomes in } A}{\text{Total number of outcomes in } S}$$

3. **Probability of the Union of Two Events:** For two events (A) and (B)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- If (A) and (B) are mutually exclusive (disjoint), then $P(A \cap B) = 0$, and the formula simplifies to

$$P(A \cup B) = P(A) + P(B)$$

4. **Probability of the Intersection of Independent Events:** If (A) and (B) are independent events (i.e., the occurrence of one does not affect the occurrence of the other), then

$$P(A \cap B) = P(A) \cdot P(B)$$

5. **Sum of Probabilities in Sample Space:** The probabilities of all individual outcomes in the sample space must add up to 1:

$$\sum P(\text{individual outcomes in } S) = 1$$

6. **Long-Run Proportions:** Probabilities are often interpreted as long-run proportions. For example:

- An event with $P(A) = 0$ never occurs.
- An event with $P(A) = 1$ occurs in every trial.
- An event with $P(A) = 0.5$ occurs in about half the trials in the long run.

1.5 Addition Law of Probability and Mutually Exclusive Events

The addition law of probability states that if two events (A) and (B) are mutually exclusive (disjoint), they cannot occur simultaneously. This means that the probability of either event (A) or event (B) occurring is simply the sum of their individual probabilities:

$$P(A \cup B) = P(A) + P(B)$$

Here,

$P(A \cap B) = 0$ because the events do not overlap.

Examples

1. **Drawing a single card:** In a deck of cards, the events of drawing a "4" and drawing a "6" are mutually exclusive, as a single card cannot be both a 4 and a 6.

$$P(4 \text{ or } 6) = P(4) + P(6) = \frac{1}{52} + \frac{1}{52} = \frac{2}{52} = \frac{1}{26}$$

2. **Rolling a die:** The events of rolling a "1" and rolling a "2" are mutually exclusive, as a single roll cannot show both outcomes.

$$P(1 \cup 2) = P(1) + P(2) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

3. **What is the probability of {H, T} by tossing a single coin?**

Simple events: Outcomes such as (H) (Heads) and (T) (Tails) when a single coin is tossed are considered elementary events.

Compound event: The event $(\{H, T\})$, which includes multiple simple events, represents the entire sample space of tossing a coin.

For a fair coin, each elementary event (Head or Tail) has an equal probability of occurring. The probability of each elementary event is:

$$P(H) = \frac{1}{2}, \quad P(T) = \frac{1}{2}$$

The compound event $(\{H, T\})$ consists of all possible outcomes in the sample space. Since the coin must result in either a Head or a Tail, the probability of $(\{H, T\})$ is the sum of the probabilities of its constituent elementary events:

$$P(H, T) = P(H) + P(T) = \frac{1}{2} + \frac{1}{2} = 1$$

Thus, the probability of $(\{H, T\})$ (i.e., either a Head or a Tail) is 1, representing a certain event because one of these outcomes will always occur when tossing a single coin.

4. What is the probability of showing {1, 2} on its face by rolling a single dice?

When rolling a single die, the sample space consists of six simple events: $\{1, 2, 3, 4, 5, 6\}$, where each outcome is equally likely.

Simple Events: Rolling a "1" and rolling a "2" are individual simple events.

Compound Event: The event $\{1, 2\}$ is a compound event, representing two outcomes from the sample space.

The probability of each simple event is $\frac{1}{6}$

For the compound event $\{1, 2\}$, we sum the probabilities of its constituent simple events:

$$P(1, 2) = P(1) + P(2) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

Thus, the probability of the die showing either a "1" or a "2" is $\frac{1}{3}$

5. What is the probability of showing an even number on its face by rolling a single dice?

When rolling a single die, the sample space is $(\{1, 2, 3, 4, 5, 6\})$.

Simple Events: The even numbers are $(\{2, 4, 6\})$, which are distinct outcomes.

Compound Event: The event of showing an even number, $(\{2, 4, 6\})$, is a compound event consisting of three simple events.

Since the probability of each simple event is $\frac{1}{6}$, the probability of the compound event is:

$$P(2, 4, 6) = P(2) + P(4) + P(6) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$$

Thus, the probability of showing an even number is $\frac{1}{2}$.

6. What is the probability of showing an odd number on its face by rolling a single dice?

When rolling a single die, the sample space is $(\{1, 2, 3, 4, 5, 6\})$.

Simple Events: The odd numbers are $(\{1, 3, 5\})$, which are distinct outcomes.

Compound Event: The event of showing an odd number, $(\{1, 3, 5\})$, is a compound event consisting of three simple events.

Since the probability of each simple event is $(\frac{1}{6})$, the probability of the compound event is:

$$P(1, 3, 5) = P(1) + P(3) + P(5) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$$

Thus, the probability of showing an odd number is $\frac{1}{2}$.

7. What is the probability of showing a number greater than 3 on its face by rolling a single dice?

When rolling a single die, the sample space is $(\{1, 2, 3, 4, 5, 6\})$.

Simple Events: The outcomes greater than 3 are $(\{4, 5, 6\})$, which are distinct events.

Compound Event: The event of showing a number greater than 3, $(\{4, 5, 6\})$, is a compound event combining these simple events.

Each simple event has a probability of $(\frac{1}{6})$. The probability of the compound event is:

$$P(4, 5, 6) = P(4) + P(5) + P(6) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$$

Thus, the probability of rolling a number greater than 3 is $\frac{1}{2}$.

8. What is the probability of showing a number greater than 3 or an even number on its face by rolling a single dice?

When rolling a single die, the sample space is $(\{1, 2, 3, 4, 5, 6\})$.

Simple Events: The outcomes greater than 3 or even number are $(\{4, 5, 6\})$ or $(\{2, 4, 6\})$,

Compound Event: The event of showing a number greater than 3, $\{4, 5, 6\}$, and even numbers, $\{2, 4, 6\}$ are compound events combining these simple events.

Each simple event has a probability of $\frac{1}{6}$. The probability of the compound event is:

$$P(4, 5, 6 \text{ or } 2, 4, 6) = P(4) + P(5) + P(6) + P(2) + P(4) + P(6)$$

$$P(4, 5, 6 \text{ or } 2, 4, 6) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6}$$

$$P(4, 5, 6 \text{ or } 2, 4, 6) = \frac{3}{6} + \frac{3}{6} = 1$$

The assumption that the events (A: **number greater than 3**) and (B: **even number**) are mutually exclusive is wrong. The events actually overlap, as numbers like (4) and (6) are both greater than 3 and even. Ignoring the overlap $(A \cap B)$ leads to double-counting and an overestimated probability.

The correct approach uses the general addition rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If we neglect the overlap and simply sum $P(A) + P(B)$, we overestimate the probability.

The failure cases for non-mutually exclusive events demonstrate the need for a more general rule. The general addition law accounts for overlap and provides an accurate calculation of probabilities, which can handle both mutually exclusive and non mutually exclusive events.

1.5.2 General Addition Law of Probability

The general addition law of probability provides a reliable method for calculating probabilities in scenarios involving both mutually exclusive and non-mutually exclusive events. This law ensures the accurate evaluation of probabilities by accounting for any overlap between the events.

For any two events (A) and (B):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- If the events are mutually exclusive, they cannot occur simultaneously, meaning $P(A \cap B) = 0$. In this case, the formula simplifies to:

$$P(A \cup B) = P(A) + P(B)$$

- If the events are non-mutually exclusive, they can occur together (i.e., $A \cap B \neq \emptyset$). Adding $P(A)$ and $P(B)$ without adjustment would double-count the outcomes in the intersection $A \cap B$. To address this, the intersection $P(A \cap B)$ is subtracted, yielding

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

This approach ensures that probabilities are not overestimated by avoiding duplication of common outcomes. The general addition law is thus essential for correctly handling both types of events. The general addition law applies to all scenarios, whether the events are mutually exclusive or not, making it a comprehensive tool for probability calculations.

9. What is the probability of showing a number greater than 3 or an even number on its face by rolling a single dice?

Sample Space: The sample space for rolling a single die is $\{1, 2, 3, 4, 5, 6\}$.

Events:

(A): Rolling a number greater than 3 $\{4, 5, 6\}$.

(B): Rolling an even number $\{2, 4, 6\}$.

Are Events Mutually Exclusive?: No, events (A) and (B) are not mutually exclusive since they overlap $A \cap B = 4, 6$.

Simple and Compound Events:

Simple events: Outcomes like 2, 4, 5, or 6.

Compound event: The union $A \cup B$, which combines all outcomes in (A) and (B) $\{2, 4, 5, 6\}$.

Probability: Using the general addition rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A) = \frac{3}{6} = \frac{1}{2}$$

$$P(B) = \frac{3}{6} = \frac{1}{2}$$

$$P(A \cap B) = \frac{2}{6} = \frac{1}{3}$$

Substituting:

$$P(A \cup B) = \frac{1}{2} + \frac{1}{2} - \frac{1}{3} = \frac{4}{6} = \frac{2}{3}$$

The probability of rolling a number greater than 3 or an even number is $\frac{2}{3}$.

10. What is the probability of showing a number greater than 3 or an odd number on its face by rolling a single dice?

(A): The event of rolling a number greater than 3 ($\{4, 5, 6\}$).

(B): The event of rolling an odd number ($\{1, 3, 5\}$).

Sample Space: The outcomes are ($\{1, 2, 3, 4, 5, 6\}$).

Are Events Mutually Exclusive?: No, (A) and (B) overlap, as the outcome "5" satisfies both conditions $A \cap B = 5$.

Simple and Compound Events:

Simple events: Outcomes like 1, 3, 4, 5, or 6.

Compound event: The union ($A \cup B$), which includes all outcomes in (A) or (B): $\{1, 3, 4, 5, 6\}$.

Using the general addition rule for probability:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

greater than 3: $\{4, 5, 6\}$ $P(A) = \frac{3}{6} = \frac{1}{2}$.

odd: $\{1, 3, 5\}$ $P(B) = \frac{3}{6} = \frac{1}{2}$.

overlap: $\{5\}$ $P(A \cap B) = \frac{1}{6}$.

Substituting:

$$P(A \cup B) = \frac{1}{2} + \frac{1}{2} - \frac{1}{6}$$

$$P(A \cup B) = \frac{3}{6} + \frac{3}{6} - \frac{1}{6} = \frac{5}{6}$$

The probability of rolling a number greater than 3 or an odd number is $\frac{5}{6}$.

1.5.3 Completely Exhaustive Events

Completely exhaustive events are a set of events that together encompass all possible outcomes in the sample space.

At least one of these events must occur.

Example:

When rolling a single die, the possible outcomes ($\{1, 2, 3, 4, 5, 6\}$) form a completely exhaustive set of events, as every possible result of the roll is included.

1.5.4 Completely Exhaustive and Mutually Exclusive Events

If the events in a completely exhaustive set are also **mutually exclusive**, it means that no two events in the set can occur simultaneously, and together they still cover the entire sample space.

Example:

For a single die roll:

- Define events as $A = \{1\}$, $B = \{2\}$, $C = \{3\}$, $D = \{4\}$, $E = \{5\}$, $F = \{6\}$.
- These events are mutually exclusive (e.g., rolling a "1" excludes rolling any other number).
- Combined, $A \cup B \cup C \cup D \cup E \cup F = 1, 2, 3, 4, 5, 6$, they form a completely exhaustive set.

Relation to the Complement Law

The complement law states: $P(A^c) = 1 - P(A)$ Where A^c is the complement of event (A). This law highlights that (A) and A^c together form a **mutually exclusive and completely exhaustive set**:

- **Mutually Exclusive:** $A \cap A^c = \emptyset$ (they cannot occur simultaneously).
- **Completely Exhaustive:** $A \cup A^c = S$ (the sample space).

In the dice example, the event (A) (rolling a number greater than 3: $\{4, 5, 6\}$) and its complement A^c (rolling a number not greater than 3: $\{1, 2, 3\}$) are mutually exclusive and completely exhaustive, as they account for all possible outcomes and do not overlap.

11. What is the probability of showing an even number or an odd number on its face by rolling a single dice?

When rolling a single die, the possible outcomes are ($\{1, 2, 3, 4, 5, 6\}$).

Events

(A): The event of rolling an even number ($\{2, 4, 6\}$).

(B): The event of rolling an odd number ($\{1, 3, 5\}$).

Mutually Exclusive: Events (A) and (B) are mutually exclusive because they cannot occur simultaneously.

A number cannot be both even and odd at the same time $A \cap B = \emptyset$.

Completely Exhaustive: Events (A) and (B) are completely exhaustive as they together include all possible outcomes of the roll $A \cup B = 1, 2, 3, 4, 5, 6$.

Since (A) and (B) are mutually exclusive and completely exhaustive, their probabilities can be added directly:

$$P(A \cup B) = P(A) + P(B)$$

even numbers: $P(A) = \frac{3}{6} = \frac{1}{2}$.

odd numbers: $P(B) = \frac{3}{6} = \frac{1}{2}$.

Substituting:

$$P(A \cup B) = \frac{1}{2} + \frac{1}{2} = 1$$

The probability of rolling either an even number or an odd number is 1, as these events are mutually exclusive and completely exhaustive, covering the entire sample space.

12. What is the probability of showing an even number and an odd number on its face by rolling a single dice?

It is not possible for a single roll of a die to show both an even number and an odd number simultaneously. These events are mutually exclusive, meaning the occurrence of one event excludes the possibility of the other.

Since no outcome in the sample space ($\{1, 2, 3, 4, 5, 6\}$) can be both even and odd, the intersection of the events $A \cap B$ is empty:

$$P(A \cap B) = 0$$

Thus, the probability of rolling both an even number and an odd number in a single roll is 0.

2. Marginal and Joint Probability

2.1 Multiplication Law of Probability for Independent Events

The multiplication law states that if two events are independent, the probability of their joint occurrence is the product of their individual probabilities:

$$P(A \cap B) = P(A) \cdot P(B)$$

This applies whether the events occur together (simultaneously) or sequentially, as independence ensures that the occurrence of one event does not affect the other.

Independent events are events where the occurrence of one event does not affect the occurrence of the other. For example:

- Rolling a die and flipping a coin are independent events. The outcome of the coin flip has no impact on the die roll.

Examples

1. Coin Flip and Die Roll:

Let (A): Flipping heads $P(A) = \frac{1}{2}$.

Let (B): Rolling a 6 on a die $P(B) = \frac{1}{6}$.

Since (A) and (B) are independent:

$$P(A \cap B) = P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$$

2. Drawing Cards:

If (A): Drawing a spade from a deck $P(A) = \frac{13}{52} = \frac{1}{4}$

and (B): Drawing a face card $P(B) = \frac{12}{52} = \frac{3}{13}$, and the two draws are independent (e.g., the card is replaced in the deck):

$$P(A \cap B) = P(A) \cdot P(B) = \frac{1}{4} \cdot \frac{3}{13} = \frac{3}{52}$$

This law applies only if the events are independent. If (A) and (B) are not independent, their probabilities must account for the dependence between them. The multiplication law is a foundational rule in probability and is widely used in scenarios involving independent random events.

Examples

1. What is the probability of two heads by tossing two coins?

The sample space for tossing two coins is ({HH, HT, TH, TT}).

Define (A): Tossing heads on the first coin $P(A) = \frac{1}{2}$.

Define (B): Tossing heads on the second coin $P(B) = \frac{1}{2}$.

The two events, (A) and (B), are independent because the outcome of one coin does not affect the other.

Using the multiplication law for independent events: $P(A \cap B) = P(A) \cdot P(B)$.

Substitute: $P(A \cap B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

Therefore, the probability of tossing two heads is $\frac{1}{4}$.

2. What is the probability of two tails by tossing two coins?

The sample space for tossing two coins is $(\{HH, HT, TH, TT\})$.

Define (A): Tossing tails on the first coin $P(A) = \frac{1}{2}$.

Define (B): Tossing tails on the second coin $P(B) = \frac{1}{2}$.

The two events, (A) and (B), are independent because the outcome of one coin does not affect the other.

Using the multiplication law for independent events: $P(A \cap B) = P(A) \cdot P(B)$.

Substitute: $P(A \cap B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

Therefore, the probability of tossing two tails is $\frac{1}{4}$.

3. What is the probability of head and tails by tossing two coins?

(A): The event of getting a head on the first coin $P(A) = \frac{1}{2}$.

(B): The event of getting a tail on the second coin $P(B) = \frac{1}{2}$.

(C): The event of getting a tail on the first coin $P(C) = \frac{1}{2}$.

(D): The event of getting a head on the second coin $P(D) = \frac{1}{2}$.

Define Two Independent Scenarios

Scenario 1: First coin shows heads and second coin shows tails $A \cap B$.

Using the multiplication law for independent events:

$$P(A \cap B) = P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

Scenario 2: First coin shows tails and second coin shows heads $C \cap D$.

Using the multiplication law for independent events:

$$P(C \cap D) = P(C) \cdot P(D) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

Combine Using the Addition Law

Since these two scenarios $A \cap B$ and $C \cap D$ are mutually exclusive (they cannot occur together), we use the addition law:

$$P((A \cap B) \cup (C \cap D)) = P(A \cap B) + P(C \cap D)$$

Substitute the values:

$$P((A \cap B) \cup (C \cap D)) = \frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$$

The probability of tossing one head and one tail (in any order) is $\frac{1}{2}$.

- 4. What is the probability of showing an even number and a number less than 4 on its face by rolling a single dice?**

Using the Multiplication Law of Probability, let's assume independence between the two events:

$$(A): \text{Rolling an even number } (\{2, 4, 6\}). \quad P(A) = \frac{3}{6} = \frac{1}{2}.$$

$$(B): \text{Rolling a number less than 4 } (\{1, 2, 3\}). \quad P(B) = \frac{3}{6} = \frac{1}{2}.$$

Assuming (A) and (B) are independent, the probability of both occurring is calculated as:

$$P(A \cap B) = P(A) \cdot P(B)$$

Substituting:

$$P(A \cap B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

The assumption of independence is incorrect because events (A) and (B) overlap. Specifically:

$A \cap B$: Rolling a number that is even and less than 4 includes only $\{2\}$, which has 1 favorable outcome.

The true probability should account for the intersection of (A) and (B), not assume independence.

Correct Approach

The events (A) and (B) are not independent because they overlap at outcome "2". To account for this dependence, we use conditional probability.

The probability of rolling a number less than 4, given that the number is even $P(B|A)$, considers only the even numbers ($\{2, 4, 6\}$). Among these, only "2" is less than 4.

$$P(B|A) = \frac{1}{3}$$

Using the multiplication law for dependent events:

$$P(A \cap B) = P(A) \cdot P(B|A)$$

Substitute the values:

$$P(A \cap B) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$$

Thus, the correct probability of rolling an even number and a number less than 4 is $\frac{1}{6}$.

The error arises from assuming that (A) and (B) are independent when they actually overlap. Independence should not be assumed unless explicitly known or proven; here, accounting for the actual intersection $A \cap B$ gives the correct result.

2.1.1 General Multiplication Law of Probability

The General Multiplication Law of Probability states that the probability of two events (A) and (B) occurring together (intersection $A \cap B$) is given by:

$$P(A \cap B) = P(A) \cdot P(B|A)$$

Where:

- $(P(B|A))$ is the conditional probability of (B) occurring, given that (A) has already occurred.

This law is applicable for both independent and dependent events:

Independent Events

The occurrence of (A) does not affect the occurrence of (B). Hence,

$$P(B|A) = P(B)$$

For independent events:

$$P(A \cap B) = P(A) \cdot P(B)$$

2.2 Dependent Events

The occurrence of (A) affects the likelihood of (B). Hence, $P(B|A)$ must be calculated explicitly.

Examples

Independent Events

Scenario: Tossing two coins.

- (A): Tossing heads on the first coin ($P(A) = \frac{1}{2}$).
- (B): Tossing heads on the second coin ($P(B) = \frac{1}{2}$).

Since the outcomes of the two coin tosses are independent, the general multiplication rule simplifies to:

$$P(A \cap B) = P(A) \cdot P(B)$$

Substitute:

$$P(A \cap B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

The probability of tossing heads on both coins is $\frac{1}{4}$.

Dependent Events

Scenario: Drawing two cards sequentially without replacement from a deck of 52 cards.

- (A): Drawing a King first $P(A) = \frac{4}{52} = \frac{1}{13}$.
- (B): Drawing a King second $P(B|A) = \frac{3}{51}$, since one King has already been removed from the deck.

Since the two draws are dependent, the general multiplication law applies:

$$P(A \cap B) = P(A) \cdot P(B|A)$$

Substitute:

$$P(A \cap B) = \frac{1}{13} \cdot \frac{3}{51} = \frac{3}{663} = \frac{1}{221}$$

The probability of drawing two Kings sequentially is $\frac{1}{221}$.

The general multiplication law is versatile as it accommodates both independent and dependent events:

- For independent events, $P(B|A) = P(B)$.
- For dependent events, $P(B|A)$ must account for the influence of (A).

5. What is the probability of showing an odd number and a number less than 4 on its face by rolling a single dice?

The sample space for rolling a single die is $(\{1, 2, 3, 4, 5, 6\})$.

Define (A): Rolling an odd number ($\{1, 3, 5\}$) $\rightarrow P(A) = \frac{3}{6} = \frac{1}{2}$.

Define (B): Rolling a number less than 4 ($\{1, 2, 3\}$) $\rightarrow P(B) = \frac{3}{6} = \frac{1}{2}$.

Events (A) and (B) are not independent, as they overlap at outcomes $\{1, 3\}$.

The conditional probability of rolling a number less than 4 given that the number is odd $P(B|A)$ considers the odd outcomes ($\{1, 3, 5\}$). Among these, two outcomes ($\{1, 3\}$) are less than 4.

$$P(B|A) = \frac{2}{3}.$$

Using the multiplication law for dependent events

$$P(A \cap B) = P(A) \cdot P(B|A).$$

Substitute the values

$$P(A \cap B) = \frac{1}{2} \cdot \frac{2}{3} = \frac{2}{6} = \frac{1}{3}.$$

The probability of rolling an odd number and a number less than 4 is $\frac{1}{3}$.

6. What is the probability of showing an even number and a number less than 5 on its face by rolling a single dice?

The sample space for rolling a single die is ($\{1, 2, 3, 4, 5, 6\}$).

Define (A): Rolling an even number ($\{2, 4, 6\}$) $\rightarrow P(A) = \frac{3}{6} = \frac{1}{2}$.

Define (B): Rolling a number less than 5 ($\{1, 2, 3, 4\}$) $\rightarrow P(B) = \frac{4}{6} = \frac{2}{3}$.

Events (A) and (B) overlap, as outcomes $\{2, 4\}$ satisfy both conditions $A \cap B = 2, 4$.

Conditional probability of rolling a number less than 5, given that the number is even $P(B|A)$, consider the even numbers ($\{2, 4, 6\}$). Among these, two outcomes ($\{2, 4\}$) are less than 5.

$$P(B|A) = \frac{2}{3}$$

Using the multiplication law for dependent events:

$$P(A \cap B) = P(A) \cdot P(B|A).$$

Substitute the values

$$P(A \cap B) = \frac{1}{2} \cdot \frac{2}{3} = \frac{2}{6} = \frac{1}{3}.$$

The probability of rolling an even number and a number less than 5 is $\frac{1}{3}$.

7. **What is the probability of showing an even number and a number less than 5 on its face by rolling a single dice? (BY ASSUMING INDEPENDENCE)**

The sample space for rolling a single die is $(\{1, 2, 3, 4, 5, 6\})$.

Define (A): Rolling an even number $(\{2, 4, 6\}) \rightarrow P(A) = \frac{3}{6} = \frac{1}{2}$.

Define (B): Rolling a number less than 5 $(\{1, 2, 3, 4\}) \rightarrow P(B) = \frac{4}{6} = \frac{2}{3}$.

Assuming (A) and (B) are independent, the multiplication law of probability is applied

$$P(A \cap B) = P(A) \cdot P(B)$$

Substituting the probabilities:

$$P(A \cap B) = \frac{1}{2} \cdot \frac{2}{3} = \frac{2}{6} = \frac{1}{3}$$

The probability of rolling an even number and a number less than 5 is $\frac{1}{3}$.

8. **What is the probability of showing an odd number and a number less than 3 on its face by rolling a single dice?**

Assuming Independence

The sample space is $(\{1, 2, 3, 4, 5, 6\})$.

Define (A): Rolling an odd number $(\{1, 3, 5\}) \rightarrow P(A) = \frac{3}{6} = \frac{1}{2}$.

Define (B): Rolling a number less than 3 $(\{1, 2\}) \rightarrow P(B) = \frac{2}{6} = \frac{1}{3}$.

Assuming (A) and (B) are independent, the probability of both events occurring is:

$$P(A \cap B) = P(A) \cdot P(B) \text{ Substitute : } [P(A \cap B) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}]$$

Assuming Dependence

The outcomes that satisfy $A \cap B$ (odd and less than 3) are $(\{1\})$.

Conditional probability $P(B | A)$ considers only odd outcomes $(\{1, 3, 5\})$. Among these, only "1" is less than 3. Hence:

$$P(B|A) = \frac{1}{3}$$

Using the general multiplication rule:

$$P(A \cap B) = P(A) \cdot P(B|A) \text{ Substitute : } [P(A \cap B) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}.$$

2.3 Conditional Probability

Conditional probability is the probability of an event (B) occurring, given that another event (A) has already occurred. It is denoted as $P(B | A)$, and mathematically defined as

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \quad \text{provided } P(A) > 0$$

This formula helps calculate the likelihood of (B) happening in the context of (A), and is particularly useful for **dependent events**, where the occurrence of one affects the likelihood of the other.

- It captures the influence of one event on another.
- Helps distinguish between **independence** and **dependence**.
- **Dependent Events**: Used when events are not independent (e.g., drawing cards without replacement).
- **Restricted Sample Space**: When event (A) has already occurred, the sample space reduces to include only the outcomes favorable to (A).
- Used in real-world applications like medical testing (e.g., probability of having a disease given a positive test result).

Examples

1. Dependent Events Example

Scenario: Drawing cards without replacement.

- Event (A): Drawing a red card from a deck

$$P(A) = \frac{26}{52} = \frac{1}{2}$$

- Event (B): Drawing a face card (Jack, Queen, King) on the second draw, given that the first card was red.

If (A) has occurred (first card is red), there are 25 red cards remaining, and 6 of them are face cards.

$$P(B|A) = \frac{\text{Number of Red Face Cards Remaining}}{\text{Number of Remaining Red Cards}} = \frac{6}{25}$$

Using the **general multiplication law**:

$$P(A \cap B) = P(A) \cdot P(B|A) = \frac{1}{2} \cdot \frac{6}{25} = \frac{6}{50} = \frac{3}{25}$$

2. Independent Events Example

Scenario: Flipping a coin and rolling a die.

- Event (A): Tossing heads $P(A) = \frac{1}{2}$.
- Event (B): Rolling a "6" $P(B) = \frac{1}{6}$.

Since (A) and (B) are independent,

$$P(B|A) = P(B).$$

Using the multiplication law

$$P(A \cap B) = P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$$

2.4 Bayes' Rule (OPTIONAL)

Bayes' Rule (or Bayes' Theorem) provides a way to update probabilities as new information becomes available. It links conditional probability and reasoning under uncertainty, answering the question: "Given a certain event occurred, what is the probability of a related event?"

Formula

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Where:

- $P(A | B)$: Probability of (A) given (B).
- $P(B | A)$: Probability of (B) given (A).
- $P(A)$: Prior probability of (A) (before knowing (B)).
- $P(B)$: Total probability of (B) occurring.

Bayes' Rule essentially **reverses conditional probabilities** by using existing information about ($P(B | A)$) to calculate ($P(A | B)$).

Think of Bayes' Rule as a process of **updating beliefs** based on evidence.

- **Prior Probability ($P(A)$)**: What you initially believed (before observing (B)).
- **Likelihood $P(B | A)$:** How likely the evidence (B) is if your belief (A) is true.
- **Posterior Probability $P(A | B)$** : The updated probability of (A), after considering the evidence (B).

Let's revisit the card-drawing example with **dependent events**:

- (A): The event of drawing a face card (Jack, Queen, King).
- (B): The event of drawing a red card.

Question: If the first card drawn is red ((B)), what is the probability it is a face card ((A))?

1. $P(A)$: Prior probability of drawing a face card = $\frac{12}{52}$.
2. $P(B | A)$: Probability of drawing a red card if it's a face card = $\frac{6}{12}$.
3. $P(B)$: Total probability of drawing a red card = $\frac{26}{52}$.

Apply Bayes' Rule:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Substitute:

$$P(A|B) = \frac{\frac{6}{12} \cdot \frac{12}{52}}{\frac{26}{52}} = \frac{\frac{72}{624}}{\frac{26}{52}} = \frac{6}{26} = \frac{3}{13}$$

After observing a red card, the probability it is a face card is $\frac{3}{13}$.

Bayes' Rule is widely applied in practical scenarios:

1. **Medical Diagnosis:** Probability of having a disease given a positive test result.
2. **Spam Detection:** Probability an email is spam given certain keywords.
3. **Predictive Analytics:** Update probabilities as new data emerges.

Bayes' Rule combines prior knowledge and new evidence to refine probabilities, making it invaluable for decision-making under uncertainty.

2.5 Marginal and Joint Probabilities

Marginal probability represents the likelihood of a single event (E) occurring, irrespective of any other events. It is denoted as $P(E)$ and is calculated by dividing a specific subset of outcomes by the total number of possible outcomes in the sample space.

Formula:

$$P(E) = \frac{\text{Number of favorable outcomes for } E}{\text{Total number of possible outcomes}}$$

Example 1: The probability that a person owns a Ford car can be calculated as:

$$P(\text{Ford}) = \frac{\text{Number of Ford owners}}{\text{Total number of car owners}}$$

Example 2: The probability that a person wears glasses

$$P(\text{Glasses}) = \frac{\text{Number of people wearing glasses}}{\text{Total number of people}}$$

Joint probability is the likelihood of two events E_1 and E_2 occurring **simultaneously**. It is also called the probability of the **intersection** of events, denoted as $P(E_1 \cap E_2)$.

Formula:

$$P(E_1 \cap E_2) = \frac{\text{Number of outcomes satisfying both } E_1 \text{ and } E_2}{\text{Total number of possible outcomes}}$$

For **dependent events**, the general multiplication law is applied:

$$P(E_1 \cap E_2) = P(E_1) \cdot P(E_2|E_1)$$

For **independent events**, it simplifies to

$$P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$$

Example 1: The probability of a person owning **both a Ford and a Chevrolet**:

$$P(\text{Ford and Chevrolet}) = \frac{\text{Number of people owning both cars}}{\text{Total number of car owners}}$$

Example 2: The probability of a person being **a redhead and wearing glasses**:

$$P(\text{Redhead and Glasses}) = \frac{\text{Number of redheads who wear glasses}}{\text{Total number of people}}$$

3. Association between two categorical variables

When analyzing the association between two categorical variables, we often examine how the distribution of one variable depends on the categories of another. This can be done using tools such as tables, bar graphs, or pie charts to summarize the percentage distribution for one variable. To delve deeper into the relationship, we assess how these percentages vary based on categories of the other variable.

A contingency table is a structured display of two categorical variables, with rows representing the categories of one variable and columns representing those of the other. Each cell in the table contains the count of observations corresponding to a particular pair of categories. This approach is useful for identifying relationships and patterns between the two variables.

When the data is sufficiently large, proportions derived from such summaries can be converted to probabilities using the frequentist approach to probability. This approach interprets probability as the relative frequency of an event occurring over repeated trials or observations. For instance, if the proportion of a particular category combination is 0.25, this can be interpreted as the probability of that category combination occurring, assuming a large sample size and that the data represents the underlying population effectively.

By using the relative frequency approximation, proportions serve as an empirical measure of probability, reflecting the likelihood of an observation belonging to a specific category combination within the scope of the data. This enables us to represent associations and relationships between the two variables not only as proportions but also as probabilities.

Example

To examine the relationship between two categorical variables, we will utilize the "Play Tennis" dataset. This dataset serves as an ideal tool for learning due to its simplicity and ease of comprehension. It provides a concise collection of observations where various categorical attributes, such as weather conditions, temperature, humidity, and wind, influence the decision to play tennis on a given day. By analyzing this dataset, we can illustrate how to identify associations between categorical variables effectively. Its straightforward structure makes it particularly suitable for introducing fundamental concepts in data analysis.

3.1 Create a Contingency Table using python

Understand the Dataset: Identify the categorical variables in your dataset that you wish to analyze. For example, in the "Play Tennis" dataset, variables include Outlook, Temperature, Humidity, Wind, and Play Tennis.

The "Play Tennis" dataset is a commonly used dataset in machine learning and statistics for classification and decision-making exercises. It contains attributes that affect whether or not tennis is played on a given day. The categorical attributes include:

- Outlook: Weather conditions (e.g., Sunny, Overcast, Rain).

- Temperature: Temperature levels (e.g., Hot, Mild, Cool).
- Humidity: Humidity levels (e.g., High, Normal).
- Wind: Wind conditions (e.g., Weak, Strong).
- Play Tennis: Decision outcome (e.g., Yes or No).

The dataset helps analyze how weather conditions influence the decision to play tennis. For instance, we can assess how different categories of Outlook relate to the Play Tennis decision.

To create a contingency table, begin by identifying two categorical variables from your dataset that you wish to analyze. These variables should have distinct categories (e.g., "Sunny" and "Rainy" for weather conditions or "Yes" and "No" for a decision outcome). The goal is to explore the association between these variables by summarizing their relationship in a tabular format.

A contingency table is a structured display that organizes data into rows and columns, where:

- Each row corresponds to one category of the first variable.
- Each column represents one category of the second variable.
- The cells within the table contain the frequency or count of observations for each combination of categories.

By constructing the contingency table, you can observe how the categories of one variable are distributed across the categories of the other. This method is essential for understanding potential relationships or patterns between the two variables. For example, in a dataset on weather and activities, you could analyze how weather conditions (variable 1) influence the likelihood of engaging in an outdoor activity (variable 2).

```
import pandas as pd

# Define the "Play Tennis" dataset
data = {
    'Outlook': ['Sunny', 'Sunny', 'Overcast', 'Rain', 'Rain', 'Rain', 'Overcast',
               'Sunny', 'Sunny', 'Rain', 'Sunny', 'Overcast', 'Overcast', 'Rain'],
    'Play Tennis': ['No', 'No', 'Yes', 'Yes', 'Yes', 'No', 'Yes', 'No', 'Yes',
                   'Yes', 'Yes', 'Yes', 'Yes', 'No']
}
```

```

}

# Convert the dataset into a pandas DataFrame
df = pd.DataFrame(data)

# Generate a contingency table (Outlook vs Play Tennis)
contingency_table = pd.crosstab(df['Outlook'], df['Play Tennis'])
print("Contingency Table (Outlook vs Play Tennis):")
print(contingency_table)

```

Output

```

Contingency Table (Outlook vs Play Tennis):
Play Tennis  No  Yes
Outlook
Overcast      0   4
Rain          2   3
Sunny         3   2

```

```

# Add row/column totals
ct_margins = pd.crosstab(df['Outlook'], df['Play Tennis'], margins=True)
print("\nContingency Table with Margins:")
print(ct_margins)

```

Output

```

Contingency Table (Outlook vs Play Tennis)
Play Tennis  No  Yes  All
Outlook
Overcast      0   4    4
Rain          2   3    5
Sunny         3   2    5
All           5   9   14

```

Probability Analysis Using Contingency Tables

1. Joint Probabilities:

Joint probabilities measure the likelihood of observing a specific combination of categories.

- Computation: Divide each cell frequency by the total dataset size.

```

# Calculate Joint Probabilities
joint_probabilities = pd.crosstab(df['Outlook'], df['Play Tennis'], normalize=True)
print("\nJoint Probabilities (Outlook and Play Tennis):")
print(joint_probabilities)

```

Output

Joint Probabilities (Outlook and Play Tennis):

Play Tennis	No	Yes
Outlook		
Overcast	0.000000	0.285714
Rain	0.142857	0.214286
Sunny	0.214286	0.142857

Observations:

- Joint probabilities highlight the likelihood of combinations like Sunny & No or Rain & Yes.
- Higher probabilities indicate stronger associations.

2. Marginal Probabilities:

Marginal probabilities represent the overall likelihood of observing a specific category of a single variable.

- Computation: Divide the row/column totals by the total dataset size.

```
# Find the Marginal Probabilities
marginal_probabilities = pd.crosstab(df['Outlook'], df['Play Tennis'],
normalize=True, margins=True)
print("\nMarginal Probabilities:")
print(marginal_probabilities)
```

Output

Contingency Table (Outlook vs Play Tennis): Relative frequencies

Play Tennis	No	Yes	All
Outlook			
Overcast	0.000000	0.285714	0.285714
Rain	0.142857	0.214286	0.357143
Sunny	0.214286	0.142857	0.357143
All	0.357143	0.642857	1.000000

Observations:

- Marginal probabilities for Outlook categories (Sunny, Overcast, Rain) show the distribution of weather conditions.
- Marginal probabilities for Play Tennis (Yes, No) reflect the overall likelihood of playing or not playing tennis.

3. Conditional Probabilities:

Conditional probabilities measure the probability of one variable given a specific category of another variable.

- Computation Examples:

- $P(\text{Outlook} = \text{Sunny} \mid \text{Play Tennis} = \text{Yes})$
- $P(\text{Play Tennis} = \text{Yes} \mid \text{Outlook} = \text{Sunny})$

```
# Conditional Probabilities: P(Outlook | Play Tennis)
conditional_probabilities_ogpt = marginal_probabilities.apply(lambda x: x /
x[-1])[:-1]
print("\nConditional Probabilities (Outlook given Play Tennis):")
print(conditional_probabilities_ogpt)
```

Output

```
Conditional Probabilities (Outlook given Play Tennis):
Play Tennis   No      Yes      All
Outlook
Overcast      0.0    0.444444  0.285714
Rain          0.4    0.333333  0.357143
Sunny         0.6    0.222222  0.357143
All           1.0    1.000000  1.000000
```

```
# Conditional Probabilities: P(Play Tennis | Outlook)
conditional_probabilities_ptgo = marginal_probabilities.apply(lambda x: x / x[-1],
axis=1)[:-1]
print("\nConditional Probabilities (Play Tennis given Outlook):")
print(conditional_probabilities_ptgo)
```

Output

```
Conditional Probabilities (Play Tennis given Outlook):
Play Tennis      No      Yes  All
Outlook
Overcast      0.000000  1.000000  1.0
Rain          0.400000  0.600000  1.0
Sunny         0.600000  0.400000  1.0
All           0.357143  0.642857  1.0
```

Observations:

- Conditional probabilities for Play Tennis = Yes are not uniform across Outlook categories.
- Significant variations between conditional and marginal probabilities indicate dependency.

3.2 Independence Analysis

- Method: Compare conditional probabilities to marginal probabilities:
 - Independent: Conditional probabilities equal marginal probabilities.
 - Dependent: Conditional probabilities vary from marginal probabilities.

```
# Compare Conditional and Marginal Probabilities
print("\nComparison of Conditional to Marginal Probabilities:")
for outlook in conditional_probabilities_ptgo.index:
    for play_tennis in conditional_probabilities_ptgo.columns:
        cond_prob = conditional_probabilities_ptgo.loc[outlook, play_tennis]
        marg_prob = marginal_probabilities.loc['All', play_tennis]
        print(f"P({play_tennis} | {outlook}) = {cond_prob:.2f}, Marginal
P({play_tennis}) = {marg_prob:.2f}")
        print(f"Difference: {abs(cond_prob - marg_prob):.2f}")
```

Output

```
Comparison of Conditional Probabilities (Play Tennis given Outlook) to Marginal
Probabilities:
P(No | Overcast) = 0.00, Marginal P(No) = 0.64
Difference: 0.64
P(Yes | Overcast) = 1.00, Marginal P(Yes) = 0.64
Difference: 0.36
P(No | Rain) = 0.40, Marginal P(No) = 0.64
Difference: 0.24
P(Yes | Rain) = 0.60, Marginal P(Yes) = 0.64
Difference: 0.04
P(No | Sunny) = 0.60, Marginal P(No) = 0.64
Difference: 0.04
P(Yes | Sunny) = 0.40, Marginal P(Yes) = 0.64
Difference: 0.24
```

Observations:

- Dependencies exist if conditional probabilities deviate significantly from marginal probabilities. For example, Outlook = Overcast often leads to Play Tennis = Yes.

Visualization: Stacked Bar Plot

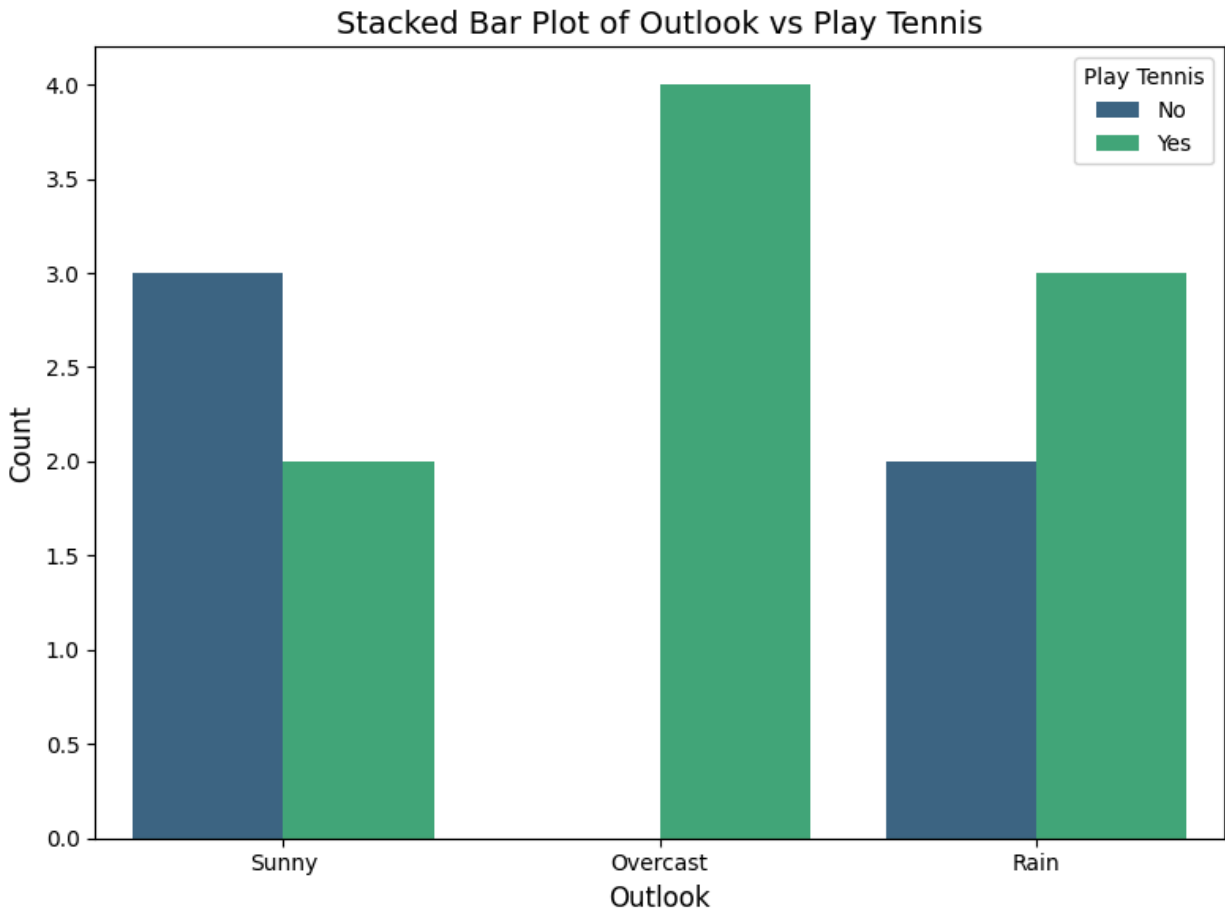
Visualize the relationship between Outlook and Play Tennis outcomes using a stacked bar chart.

```
import seaborn as sns
import matplotlib.pyplot as plt

# Create Stacked Bar Plot
plt.figure(figsize=(8, 6))
sns.countplot(data=df, x='Outlook', hue='Play Tennis', palette='viridis')
plt.title('Outlook vs Play Tennis', fontsize=14)
plt.xlabel('Outlook', fontsize=12)
plt.ylabel('Count', fontsize=12)
plt.legend(title='Play Tennis')
```

```
plt.tight_layout()
plt.show()
```

Output



Observations:

- The distribution of Yes and No outcomes varies significantly across Outlook categories.
- A higher proportion of Yes under Overcast suggests its favorable impact on tennis-playing decisions.

3.3 Casestudy 2: Tips Dataset

Introduction to the Tips Dataset

The tips dataset is a well-known dataset in data visualization and statistical analysis, frequently used in Python libraries like seaborn and pandas. It provides information on customer behavior at a restaurant, including details about bills, tips, and demographics. The dataset allows us to analyze relationships and patterns, such as tipping behavior and its dependence on gender, smoking habits, and other variables.

Features in the Dataset

1. **total_bill:**
 - The total bill amount paid by the customer, including food and drinks.
 - A numeric feature used to study spending patterns.
2. **tip:**
 - The tip amount left by the customer.
 - A numeric feature for analyzing tipping behavior in relation to other factors.
3. **sex:**
 - The gender of the customer (Male or Female).
 - A categorical feature to study behavioral differences by gender.
4. **smoker:**
 - Indicates whether the customer is a smoker (Yes or No).
 - A categorical feature for analyzing the influence of smoking habits.
5. **day:**
 - The day of the week (Thurs, Fri, Sat, Sun) when the customer visited.
 - A categorical feature to explore variations in behavior across weekdays.
6. **time:**
 - The time of day (Lunch or Dinner) the customer visited.
 - Useful for analyzing how behavior changes across meal times.
7. **size:**
 - The size of the customer's dining party.
 - A numeric feature to study how group size affects tipping behavior.

Importing the Dataset

- The dataset is accessed directly from the seaborn library and loaded into a pandas DataFrame for further analysis.

Code:

```
# Load the `tips` dataset
```

```
tips = sns.load_dataset('tips')
tips.head()
```

Output

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

Creating a Contingency Table

A contingency table summarizes the frequency of combinations of two categorical variables. Here, we analyze the relationship between **sex** and **smoker**.

Code:

```
# Generate a Contingency Table
contingency_table = pd.crosstab(tips['sex'], tips['smoker'])
print("Contingency Table (Sex vs Smoker):")
print(contingency_table)
```

Output

Contingency Table (Sex vs Smoker):		
smoker	Yes	No
sex		
Male	60	97
Female	33	54

Observations:

- The contingency table provides a count of smokers and non-smokers for each gender (**Male** and **Female**).
- It highlights the distribution of smoking behavior across genders.

Performing Probability Analysis

Joint Probabilities

Joint probabilities measure the likelihood of observing a specific combination of categories (**sex** and **smoker**).

Code:

```
# Calculate Joint Probabilities
joint_probabilities = pd.crosstab(tips['sex'], tips['smoker'], normalize=True)
```

```
print("\nJoint Probabilities (Sex and Smoker):")
print(joint_probabilities)
```

Output

```
Joint Probabilities (Sex and Smoker):
smoker      Yes      No
sex
Male    0.245902  0.397541
Female  0.135246  0.221311
```

Observations:

- Joint probabilities show the relative frequency of each combination (e.g., Male & Smoker, Female & Non-Smoker).
- They highlight how smoking behavior is distributed across genders.

Marginal Probabilities

Marginal probabilities indicate the overall likelihood of each category, irrespective of the other variable.

Code:

```
# Calculate Marginal Probabilities
marginal_probabilities = pd.crosstab(tips['sex'], tips['smoker'], normalize=True,
margins=True)
print("\nMarginal Probabilities (Sex and Smoker):")
print(marginal_probabilities)
```

Output

```
Marginal Probabilities (Sex and Smoker):
smoker      Yes      No      All
sex
Male    0.245902  0.397541  0.643443
Female  0.135246  0.221311  0.356557
All     0.381148  0.618852  1.000000
```

Observations:

- Marginal probabilities for sex reflect the proportion of males and females.
- Marginal probabilities for smoker show the likelihood of being a smoker or non-smoker, regardless of gender.

Conditional Probabilities

Conditional probabilities provide the likelihood of one category given the occurrence of another.

Code:

```
# Conditional Probabilities: Smoker given Sex
conditional_probabilities_smoker_given_sex =
contingency_table.div(contingency_table.sum(axis=1), axis=0)
print("\nConditional Probabilities (Smoker given Sex):")
print(conditional_probabilities_smoker_given_sex)

# Conditional Probabilities: Sex given Smoker
conditional_probabilities_sex_given_smoker =
contingency_table.div(contingency_table.sum(axis=0), axis=1)
print("\nConditional Probabilities (Sex given Smoker):")
print(conditional_probabilities_sex_given_smoker)
```

Output

```
Conditional Probabilities (Smoker given Sex):
smoker      Yes      No
sex
Male    0.382166  0.617834
Female  0.379310  0.620690

Conditional Probabilities (Sex given Smoker):
smoker      Yes      No
sex
Male    0.645161  0.642384
Female  0.354839  0.357616
```

Observations:

- **Smoker given Sex:**
 - These probabilities reveal smoking behavior for males vs. females.
- **Sex given Smoker:**
 - These probabilities show the gender distribution among smokers and non-smokers.

Independence Analysis

- **Goal:** To determine if **sex** and **smoker** are independent or dependent variables.
- **Method:** Compare conditional probabilities to marginal probabilities.
 - **Independence:** Conditional probabilities are equal to marginal probabilities.
 - **Dependence:** Deviations between conditional and marginal probabilities indicate dependence.

Code:

```
# Compare Conditional Probabilities to Marginal Probabilities
print("\nComparison of Conditional Probabilities to Marginal Probabilities:")
for sex in conditional_probabilities_smoker_given_sex.index:
    for smoker in conditional_probabilities_smoker_given_sex.columns:
        cond_prob = conditional_probabilities_smoker_given_sex.loc[sex, smoker]
        marg_prob = marginal_probabilities.loc['All', smoker]
        print(f"P({smoker} | {sex}) = {cond_prob:.2f}, Marginal P({smoker}) = {marg_prob:.2f}")
        print(f"Difference: {abs(cond_prob - marg_prob):.2f}")
```

Output

```
Comparison of Conditional Probabilities to Marginal Probabilities:
P(Yes | Male) = 0.38, Marginal P(Yes) = 0.38
Difference: 0.00
P(No | Male) = 0.62, Marginal P(No) = 0.62
Difference: 0.00
P(Yes | Female) = 0.38, Marginal P(Yes) = 0.38
Difference: 0.00
P(No | Female) = 0.62, Marginal P(No) = 0.62
Difference: 0.00
```

Observations:

- Significant differences between conditional and marginal probabilities suggest that **sex** and **smoker** are dependent variables.
- For example, if males are more likely to smoke than females, the two variables are not independent.

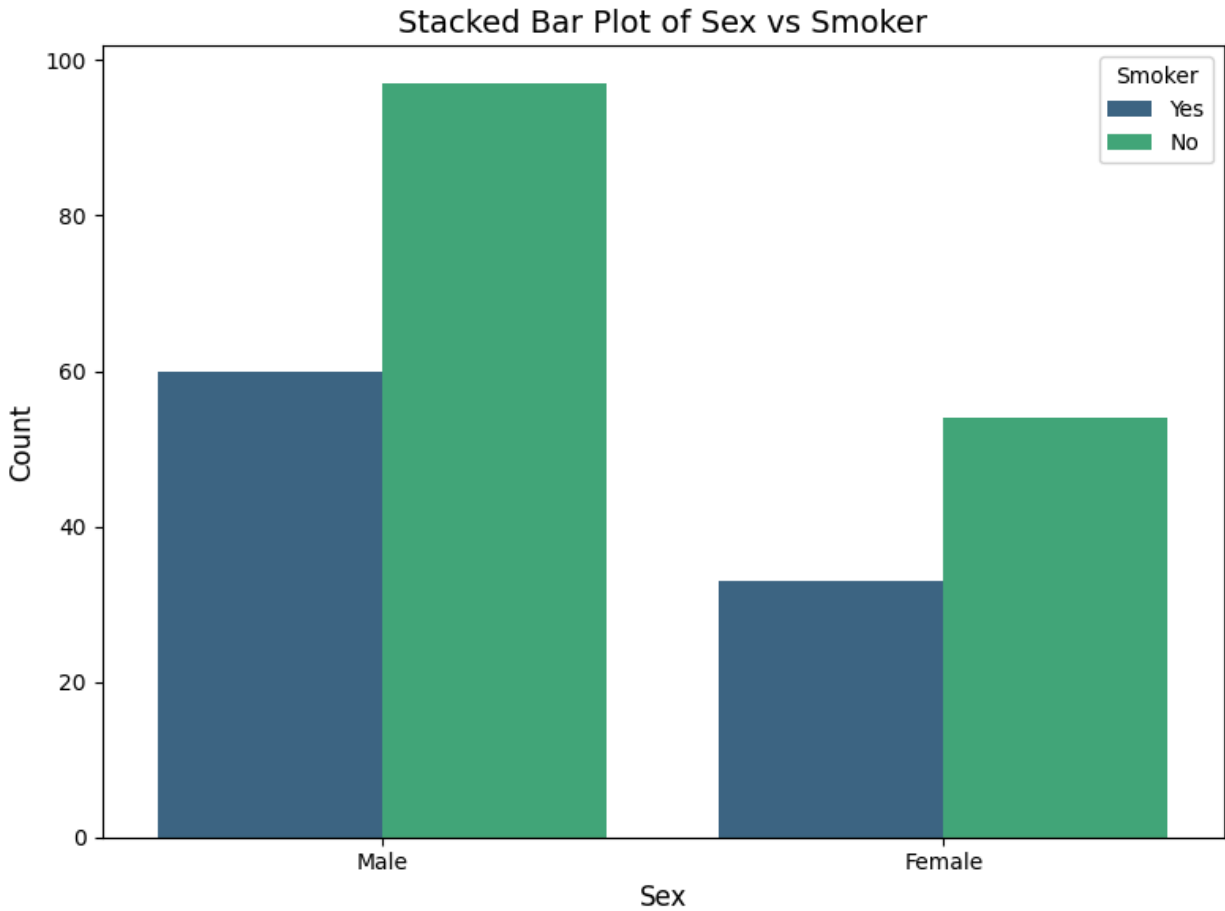
Visualization

A stacked bar plot provides a clear visual representation of the relationship between **sex** and **smoker**.

Code:

```
# Visualization using Stacked Bar Plot
plt.figure(figsize=(8, 6))
sns.countplot(data=tips, x='sex', hue='smoker', palette='viridis')
plt.title('Stacked Bar Plot of Sex vs Smoker', fontsize=14)
plt.xlabel('Sex', fontsize=12)
plt.ylabel('Count', fontsize=12)
plt.legend(title='Smoker')
plt.tight_layout()
plt.show()
```

Output



Observations:

- The bar plot illustrates the counts of smokers and non-smokers for each gender.
- It provides an intuitive understanding of how smoking behavior differs between males and females.